DSBDA Practical Viva Questions with Answers

1. Data Wrangling I

Q1. What is data wrangling?
Data wrangling is the process of cleaning, transforming, and preparing raw data into a proper format for analysis.

Q2. Why do we use pandas in Python?
Pandas is used for data handling and analysis. It provides DataFrame and Series structures to store, clean, filter, and analyze data.

Q3. What is a DataFrame?
A DataFrame is a two-dimensional table-like data structure in pandas with rows and columns.

Q4. Which function is used to load CSV data?
pd.read_csv() is used to load a CSV file into a pandas DataFrame.

Q5. What is the use of df.head()?
It displays the first five rows of the dataset by default.

Q6. What is the use of df.shape?
It shows the dimensions of the dataset in the form of rows and columns.

Q7. How do you check missing values?
We use df.isnull().sum() to check missing values column-wise.

Q8. What is the use of describe()?
describe() gives statistical information like count, mean, standard deviation, minimum, maximum, and percentiles.

Q9. What is data normalization?
Normalization is the process of scaling numeric data into a fixed range, commonly 0 to 1.

Q10. Why do we convert categorical data into numeric data?
Machine learning algorithms work on numeric values, so categorical values must be converted using label encoding or one-hot encoding.

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2. Data Wrangling II

Q1. What is data preprocessing?
Data preprocessing means preparing raw data by handling missing values, outliers, incorrect values, and inconsistent data.

Q2. What are missing values?
Missing values are blank or null values in the dataset.

Q3. How can missing values be handled?
Missing values can be handled by removing rows, filling with mean, median, mode, or using forward/backward fill.

Q4. When do we use mean to fill missing values?
Mean is used when data is normally distributed and does not contain extreme outliers.

Q5. When do we use median instead of mean?
Median is preferred when the data contains outliers.

Q6. What are outliers?
Outliers are extreme values that are very different from other values in the dataset.

Q7. How do you detect outliers?
Outliers can be detected using boxplot, Z-score, or IQR method.

Q8. What is IQR?
IQR stands for Interquartile Range.
Formula: IQR = Q3 - Q1

Q9. What is the formula to detect outliers using IQR?
Lower limit = Q1 - 1.5 * IQR
Upper limit = Q3 + 1.5 * IQR

Q10. What is data transformation?
Data transformation means changing the scale or distribution of data using techniques like log transformation, square root transformation, or normalization.

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3. Descriptive Statistics

Q1. What is descriptive statistics?
Descriptive statistics summarizes and describes the main features of data.

Q2. What is mean?
Mean is the average value of data.
Formula: sum of values / number of values

Q3. What is median?
Median is the middle value of sorted data.

Q4. What is mode?
Mode is the most frequently occurring value in the dataset.

Q5. What is standard deviation?
Standard deviation measures how much data values are spread from the mean.

Q6. What is variance?
Variance is the square of standard deviation. It shows data spread.

Q7. What is percentile?
Percentile tells the value below which a certain percentage of data falls.

Q8. What is the difference between mean and median?
Mean is affected by outliers, but median is less affected by outliers.

Q9. Why do we use groupby()?
groupby() is used to group data based on a categorical column and apply statistical functions.

Q10. Example: why group numeric variables by categorical variables?
For example, we can calculate average marks of students grouped by gender or class.

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4. Data Analytics I — Linear Regression

Q1. What is linear regression?
Linear regression is a supervised learning algorithm used to predict continuous numeric values.

Q2. Why is linear regression used in house price prediction?
Because house price is a continuous value, so linear regression can predict price using features like area, rooms, and location.

Q3. What is the dependent variable?
The dependent variable is the output variable that we want to predict. Example: house price.

Q4. What is the independent variable?
Independent variables are input features used for prediction. Example: area, number of rooms, crime rate.

Q5. What is the equation of simple linear regression?
$y = mx + c$
Where y is output, x is input, m is slope, and c is intercept.

Q6. What is training data?
Training data is used to train the machine learning model.

Q7. What is testing data?
Testing data is used to check how well the model performs on unseen data.

Q8. Why do we split data into training and testing sets?
To train the model on one part and test its performance on another part.

Q9. What is mean squared error?
MSE measures the average squared difference between actual and predicted values.

Q10. What is R^2 score?
R^2 score tells how well the regression model fits the data. A value closer to 1 means better performance.

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5. Data Analytics II — Logistic Regression

Q1. What is logistic regression?
Logistic regression is a supervised learning algorithm used for classification problems.

Q2. Why is it called regression if it is used for classification?
Because it uses a regression equation internally, but applies a sigmoid function to classify output.

Q3. What type of output does logistic regression give?
It gives probability between 0 and 1.

Q4. What is sigmoid function?
Sigmoid function converts values into a range between 0 and 1.

Q5. In Social_Network_Ads dataset, what is usually predicted?
Usually, we predict whether a user purchased a product or not based on age and estimated salary.

Q6. What is binary classification?
Binary classification means classifying data into two classes, such as Yes/No or 0/1.

Q7. What is confusion matrix?
A confusion matrix is a table used to evaluate classification model performance.

Q8. What is TP?
True Positive means the actual value is positive and the model also predicted positive.

Q9. What is TN?
True Negative means the actual value is negative and the model also predicted negative.

Q10. What is FP?
False Positive means the actual value is negative but the model predicted positive.

Q11. What is FN?
False Negative means the actual value is positive but the model predicted negative.

Q12. Formula for accuracy?
Accuracy = (TP + TN) / (TP + TN + FP + FN)

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6. Data Analytics III — Naïve Bayes

Q1. What is Naïve Bayes algorithm?
Naïve Bayes is a supervised classification algorithm based on Bayes' theorem.

Q2. Why is it called "Naïve"?
It is called naïve because it assumes that all features are independent of each other.

Q3. What is Bayes' theorem?
$P(A|B) = P(B|A) * P(A) / P(B)$

Q4. Why is Naïve Bayes used on Iris dataset?
Because Iris dataset is a classification dataset where we classify flowers into species.

Q5. What are the classes in Iris dataset?
Iris-setosa, Iris-versicolor, and Iris-virginica.

Q6. What are the features of Iris dataset?
Sepal length, sepal width, petal length, and petal width.

Q7. Which Naïve Bayes type is commonly used for Iris dataset?
Gaussian Naïve Bayes, because Iris features are continuous numeric values.

Q8. What is Gaussian Naïve Bayes?
It assumes that continuous data follows a normal distribution.

Q9. Is Naïve Bayes fast?
Yes, Naïve Bayes is simple, fast, and works well for classification.

Q10. What is precision?
Precision = TP / (TP + FP)
It tells how many predicted positives are actually positive.

Q11. What is recall?
Recall = TP / (TP + FN)
It tells how many actual positives are correctly predicted.

Q12. What is error rate?
Error Rate = (FP + FN) / Total

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7. Text Analytics

Q1. What is text analytics?
Text analytics is the process of extracting useful information from text data.

Q2. What is tokenization?
Tokenization means breaking text into smaller parts like words or sentences.

Q3. What are stop words?
Stop words are common words like “is”, “the”, “and”, “are” that usually do not add much meaning.

Q4. What is stop word removal?
It is the process of removing common unnecessary words from text.

Q5. What is stemming?
Stemming reduces a word to its root form by removing suffixes.
Example: “playing” becomes “play”.

Q6. What is lemmatization?
Lemmatization converts a word to its meaningful base form using dictionary rules.
Example: “better” becomes “good”.

Q7. Difference between stemming and lemmatization?
Stemming may produce incomplete root words, but lemmatization gives meaningful dictionary words.

Q8. What is POS tagging?
POS tagging means identifying the part of speech of each word, such as noun, verb, adjective, etc.

Q9. What is TF?
TF means Term Frequency. It tells how frequently a word appears in a document.

Q10. What is IDF?
IDF means Inverse Document Frequency. It reduces the importance of common words and increases the importance of rare words.

Q11. What is TF-IDF?
TF-IDF is a numerical representation of text that shows how important a word is in a document.

Q12. Why do we convert text into numbers?
Machine learning algorithms cannot directly understand text, so text is converted into numeric form.

8. Data Visualization I — Titanic Histogram

Q1. What is data visualization?
Data visualization is the graphical representation of data using charts and plots.

Q2. Why do we use visualization?
Visualization helps us understand patterns, trends, distributions, and outliers easily.

Q3. Which library is used for visualization in this practical?
Seaborn and Matplotlib are commonly used.

Q4. What is Seaborn?
Seaborn is a Python library used for statistical data visualization.

Q5. What is the Titanic dataset?
Titanic dataset contains passenger details like age, sex, fare, class, and survival status.

Q6. What is a histogram?
A histogram shows the distribution of numeric data.

Q7. In Titanic dataset, why do we plot fare histogram?
To understand how ticket prices were distributed among passengers.

Q8. What does x-axis show in fare histogram?
The x-axis shows fare values.

Q9. What does y-axis show in histogram?
The y-axis shows frequency or count.

Q10. What can we observe from fare histogram?
Most passengers paid lower fare, while very few passengers paid very high fare.

9. Data Visualization II — Boxplot

Q1. What is a boxplot?
A boxplot displays data distribution using minimum, Q1, median, Q3, and maximum values.

Q2. Why do we use boxplot?
Boxplot is used to identify data spread and outliers.

Q3. What does the line inside the box represent?
It represents the median.

Q4. What does the box represent?
The box represents the interquartile range, from Q1 to Q3.

Q5. What do points outside the whiskers represent?
They represent outliers.

Q6. In Titanic dataset, why plot age with respect to sex?
To compare age distribution of male and female passengers.

Q7. Why add survival information in boxplot?
To compare age distribution based on survival status.

Q8. What is hue in Seaborn?
hue is used to divide data based on another categorical variable, such as survived or not survived.

Q9. What inference can be made from Titanic age boxplot?
We can observe age distribution for male and female passengers and compare survival patterns.

Q10. What does sns.boxplot() do?
It creates a boxplot using Seaborn.

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10. Data Visualization III — Iris Dataset

Q1. What is the Iris dataset?
Iris dataset is a famous dataset containing flower measurements of three Iris species.

Q2. What are the features of Iris dataset?
Sepal length, sepal width, petal length, and petal width.

Q3. What is the target variable in Iris dataset?
The target variable is species.

Q4. What type of variables are sepal length and petal length?
They are numeric continuous variables.

Q5. What type of variable is species?
Species is a categorical variable.

Q6. Why do we create histograms for each feature?
To understand the distribution of each numeric feature.

Q7. Why do we create boxplots for Iris features?
To compare spread and identify outliers.

Q8. Which feature usually separates Iris-setosa clearly?
Petal length and petal width usually separate Iris-setosa clearly.

Q9. What is an outlier in visualization?
An outlier is a data point that is far away from the normal range of values.

Q10. What inference can we write for Iris visualization?
Petal length and petal width are more useful for distinguishing species compared to sepal features.

Common Questions External Can Ask in Any Practical

Q1. Which Python libraries are commonly used in DSBDA practicals?
Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and NLTK.

Q2. What is NumPy used for?
NumPy is used for numerical calculations and array operations.

Q3. What is Scikit-learn used for?
Scikit-learn is used for machine learning algorithms, model training, testing, and evaluation.

Q4. What is Matplotlib used for?
Matplotlib is used for plotting graphs and charts.

Q5. What is the difference between classification and regression?
Regression predicts continuous values, while classification predicts categories or classes.

Q6. Give example of regression.
House price prediction.

Q7. Give example of classification.
Spam detection, Iris flower classification, or purchased/not purchased prediction.

Q8. What is train-test split?
It divides data into training data and testing data.

Q9. Why do we use random_state?
random_state gives the same split every time, so output remains consistent.

Q10. What is model accuracy?
Accuracy tells how many predictions are correct out of total predictions.

Q11. What is overfitting?
Overfitting happens when the model performs well on training data but poorly on testing data.

Q12. What is underfitting?
Underfitting happens when the model performs poorly on both training and testing data.

Very Important Formula Revision

Accuracy:
(TP + TN) / (TP + TN + FP + FN)

Error Rate:
(FP + FN) / (TP + TN + FP + FN)

Precision:
TP / (TP + FP)

Recall:
TP / (TP + FN)

F1 Score:
2 * Precision * Recall / (Precision + Recall)

IQR:
Q3 - Q1

Outlier Range:
Lower limit = Q1 - 1.5 * IQR
Upper limit = Q3 + 1.5 * IQR

Linear Regression Equation:
y = mx + c

Bayes Theorem:
P(A|B) = P(B|A) * P(A) / P(B)

Best 10 Questions to Prepare First

1. What is data wrangling?
2. How do you handle missing values?
3. What are outliers and how do you detect them?
4. What is normalization?
5. Difference between regression and classification.
6. What is confusion matrix?
7. Explain TP, TN, FP, FN.
8. Difference between logistic regression and linear regression.
9. What is Naïve Bayes and why is it called naïve?
10. Difference between stemming and lemmatization.

Group B: Big Data Analytics Viva Questions with Answers

1. Hadoop MapReduce WordCount Practical

Q1. What is Hadoop?
Hadoop is an open-source framework used to store and process large amounts of data in a distributed manner.

Q2. What is MapReduce?
MapReduce is a programming model used in Hadoop to process big data in parallel using two phases: Map phase and Reduce phase.

Q3. What is WordCount program?
WordCount is a basic Hadoop MapReduce program that counts how many times each word appears in an input file.

Q4. What is the role of Mapper?
Mapper reads input line by line, splits it into words, and emits key-value pairs like:
word, 1

Q5. What is the role of Reducer?
Reducer takes each word and all its values, then adds them to get the final count.

Example:
java → [1,1,1] → java, 3

Q6. What is key-value pair in WordCount?
A key-value pair is the intermediate data format.
Example:
("Hadoop", 1)

Q7. What is the input to Mapper?
The input to Mapper is usually a line of text from the input file.

Q8. What is the output of Mapper?
The output of Mapper is intermediate key-value pairs like:
word, 1

Q9. What is the output of Reducer?
The output of Reducer is the final word count:
word, total_count

Q10. What is local standalone mode in Hadoop?
Local standalone mode means Hadoop runs on a single machine without using HDFS or YARN. It is mainly used for testing and learning.

Q11. Which classes are commonly used in Hadoop WordCount?
Common classes are:

Mapper, Reducer, Job, Text, IntWritable, LongWritable

Q12. Why do we use Text instead of String in Hadoop?
Hadoop uses its own optimized data types. Text is used instead of Java String for efficient serialization.

Q13. Why do we use IntWritable instead of int?
IntWritable is Hadoop’s serializable version of integer data type.

Q14. What is the use of StringTokenizer?
StringTokenizer is used to split a line into individual words.

Q15. What is the main advantage of MapReduce?
It can process huge data in parallel across multiple machines.

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2. Log File Processing using MapReduce

Q1. What is log file analysis?
Log file analysis means processing system-generated log files to extract useful information like errors, warnings, user activity, or system status.

Q2. Why do we use MapReduce for log processing?
Because log files can be very large, and MapReduce can process them efficiently in parallel.

Q3. What type of data is present in log files?
Log files may contain date, time, IP address, status code, error message, request type, and system information.

Q4. What can we find from log file processing?
We can find number of errors, most visited pages, failed requests, IP-wise access count, and system warnings.

Q5. What is the role of Mapper in log processing?
Mapper reads each log line and extracts required fields like error type, IP address, or status code.

Q6. What is the role of Reducer in log processing?
Reducer combines similar keys and calculates total count or summary.

Q7. Give one example of log file MapReduce output.
Example:

ERROR 15
WARNING 8
INFO 25

Q8. What is a system log?
A system log is a file that records events occurring in an operating system or application.

Q9. What is distributed application?
A distributed application runs on multiple machines connected in a network to process data efficiently.

Q10. What is the benefit of distributed log analysis?
It saves time and handles large log files better than a single-machine program.

Q11. What is filtering in log analysis?
Filtering means selecting only required records, such as only ERROR logs from the log file.

Q12. What is parsing?
Parsing means breaking a log line into meaningful parts like timestamp, log level, and message.

Q13. What if log format is not proper?
We handle it using conditions, exception handling, or by ignoring invalid lines.

Q14. What is the key in log analysis MapReduce?
The key depends on the requirement. It can be log level, IP address, status code, or date.

Q15. What is the value in log analysis MapReduce?
Usually the value is 1, which is used for counting occurrences.

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3. Weather Data Analysis using Hadoop

Q1. What is weather data analysis?
Weather data analysis means processing weather records to find useful values like average temperature, dew point, and wind speed.

Q2. What is the input file in weather practical?
A sample text file like sample_weather.txt containing temperature, dew point, and wind speed values.

Q3. What is the objective of this practical?
The objective is to calculate average temperature, average dew point, and average wind speed.

Q4. Which framework is used for weather data analysis?
Hadoop MapReduce is used.

Q5. What is the Mapper output in weather analysis?
Mapper extracts weather values and emits key-value pairs like:

temperature, value
dew_point, value
wind_speed, value

Q6. What is the Reducer output in weather analysis?
Reducer calculates the average for each weather parameter.

Example:

temperature 32.5
dew_point 20.4
wind_speed 12.8

Q7. Formula for average?
Average = sum of values / total number of values

Q8. Why do we use MapReduce for weather data?
Because weather data can be very large and MapReduce can process it efficiently.

Q9. What is dew point?
Dew point is the temperature at which air becomes saturated with moisture and water vapor starts condensing.

Q10. What is wind speed?
Wind speed is the speed at which air moves, usually measured in km/h or m/s.

Q11. What is temperature?

Temperature shows how hot or cold the atmosphere is.

Q12. How do you handle missing values in weather data?

We can skip invalid records or replace missing values with suitable values like mean or zero, depending on the requirement.

Q13. What is the key in weather MapReduce?

The key can be weather attribute name like temperature, dew_point, or wind_speed.

Q14. What is the value in weather MapReduce?

The value is the numeric reading of that weather attribute.

Q15. What is the final result of this practical?

The final result is the average value of temperature, dew point, and wind speed.

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4. Apache Spark using Scala

Q1. What is Apache Spark?
Apache Spark is a fast big data processing framework used for large-scale data analysis.

Q2. Why is Spark faster than Hadoop MapReduce?
Spark is faster because it uses in-memory processing, while Hadoop MapReduce writes intermediate data to disk.

Q3. What is Scala?
Scala is a programming language that supports both object-oriented and functional programming.

Q4. Why is Scala used with Spark?
Spark is written in Scala, so Scala works naturally and efficiently with Spark.

Q5. What is RDD?
RDD stands for Resilient Distributed Dataset. It is the basic data structure of Spark.

Q6. What does “resilient” mean in RDD?
Resilient means Spark can recover lost data automatically using lineage information.

Q7. What does “distributed” mean in RDD?
Distributed means data is divided and processed across multiple nodes.

Q8. What is SparkContext?
SparkContext is the entry point for using Spark features in a program.

Q9. What is transformation in Spark?
Transformation creates a new RDD from an existing RDD.

Examples:
map(), filter(), flatMap()

Q10. What is action in Spark?
Action returns a result or triggers execution.

Examples:
collect(), count(), reduce(), saveAsTextFile()

Q11. Difference between transformation and action?

Transformation	Action
Creates new RDD	Returns result
Lazy execution	Triggers execution

Transformation	Action
Example: map()	Example: collect()

Q12. What is lazy evaluation in Spark?
Lazy evaluation means Spark does not execute transformations immediately. It executes only when an action is called.

Q13. What is flatMap()?
flatMap() splits one input into multiple outputs. In WordCount, it splits lines into words.

Q14. What is map() in Spark?
map() applies a function to each element of an RDD.

Q15. What is reduceByKey()?
reduceByKey() combines values having the same key.

Example:
(word, 1) values are added to get total word count.

Common Big Data Viva Questions

Q1. What is Big Data?
Big Data means very large and complex data that cannot be handled easily using traditional systems.

Q2. What are the 5 V's of Big Data?
Volume, Velocity, Variety, Veracity, and Value.

Q3. What is Volume?
Volume means huge amount of data.

Q4. What is Velocity?
Velocity means speed at which data is generated and processed.

Q5. What is Variety?
Variety means different types of data like text, image, video, logs, and sensor data.

Q6. What is Veracity?
Veracity means correctness and trustworthiness of data.

Q7. What is Value?
Value means useful information extracted from big data.

Q8. What is HDFS?
HDFS stands for Hadoop Distributed File System. It stores large files across multiple machines.

Q9. What is NameNode?
NameNode manages metadata of files stored in HDFS.

Q10. What is DataNode?
DataNode stores actual data blocks in HDFS.

Q11. What is block in HDFS?
A block is a small part of a large file stored in HDFS.

Q12. What is replication in Hadoop?
Replication means storing multiple copies of data blocks to provide fault tolerance.

Q13. What is fault tolerance?
Fault tolerance means the system continues working even if some nodes fail.

Q14. What is YARN?
YARN stands for Yet Another Resource Negotiator. It manages resources and job scheduling in Hadoop.

Q15. Difference between Hadoop and Spark?

Hadoop MapReduce	Apache Spark
Disk-based processing	In-memory processing
Slower	Faster
Best for batch processing	Supports batch and real-time processing
Written mainly in Java	Written in Scala
More code required	Less code required

Most Important Questions to Prepare First

1. What is Hadoop?
2. What is MapReduce?
3. Explain Mapper and Reducer.
4. Explain WordCount program flow.
5. What is key-value pair?
6. What is HDFS?
7. Difference between Hadoop and Spark.
8. What is RDD?
9. What is transformation and action in Spark?
10. What is lazy evaluation?
11. What is reduceByKey()?
12. How is average calculated in weather data practical?
13. What is log file processing?
14. What are 5 V's of Big Data?
15. What is local standalone setup?

One-Minute Explanation for WordCount Practical

WordCount is a basic Hadoop MapReduce program. In this program, the input text file is read line by line by the Mapper. The Mapper splits each line into words and generates key-value pairs in the form of word and 1. Then Hadoop groups the same words together and sends them to the Reducer. The Reducer adds all the values for each word and gives the final count. For example, if Hadoop appears three times, the final output will be Hadoop 3.

One-Minute Explanation for Spark Practical

Apache Spark is a fast big data processing framework. In Spark, we use RDD to store distributed data. First, we create SparkContext, then load the input file. After that, transformations like flatMap() and map() are applied. flatMap() splits lines into words, map() converts each word into a key-value pair, and reduceByKey() adds counts of the same word. Finally, an action like collect() or saveAsTextFile() gives the output. Spark is faster than Hadoop because it uses in-memory processing.